Qn			Answers	Marks
3	(a)	(i)	Light from different parts across the slit travels paths of different lengths to a point on the screen, resulting in phase differences. OR Waves will diffract when they exit the slit and spread out as they travel from the slit to the screen. With different path lengths resulting in phase differences.	B1
			When the phase difference of the light is 0, they will interfere constructively and a maximum will be produced. When the phase difference of the light is π rad, they will interfere destructively and a minimum will be produced. OR if path difference, to be in terms of integer of wavelength.	B1
	(a)	(ii)	As θ is increased, the angular separation of the two diffraction patterns increases so they change from **unresolved to resolved / distinguishable.**	B1
			The patterns are just resolved when the central maximum of one coincides with the first minimum of the other; beyond this they can be distinguished.	B1

	(a)	**(iii)**	$\theta = \frac{\lambda}{b}$ $= \frac{590 \times 10^{-9}}{0.60 \times 10^{-3}}$	M1
			$= 9.83 \times 10^{-4}$ rad	A1
	(b)	**(i)**	For 2nd order maximum of red light, $d \sin\theta = n\lambda$ $\frac{1 \times 10^{-3}}{500} \sin\theta = 2\left(650 \times 10^{-9}\right)$	M1
			$\theta = 40.5°$	A1
		(ii)	For 3rd order maximum of violet light, $\frac{1 \times 10^{-3}}{500} \sin\theta = 3\left(350 \times 10^{-9}\right)$ $\theta = 31.7°$	A1
			Therefore since the angular position of the 3rd order maximum of violet light is less that of 2nd order maximum of red light, this means that they will overlap.	A1
	(c)		Greater angular separation (greater dispersion), making the wavelengths easier to distinguish / Lower percentage	B1
			Second-order spectrum is less intense (fainter).	B1